European Journal of Sport Science

ISSN: 1746-1391 (Print) 1536-7290 (Online) Journal homepage: https://www.tandfonline.com/loi/tejs20

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To cite this article: Johannes Grünbichler, Peter Federolf & Hannes Gatterer (2019): Workload efficiency as a new tool to describe external and internal competitive match load of a professional soccer team: A descriptive study on the relationship between pre-game training loads and relative match load, European Journal of Sport Science, DOI: 10.1080/17461391.2019.1697374

To link to this article: https://doi.org/10.1080/17461391.2019.1697374

Accepted author version posted online: 25 Nov 2019.
Published online: 11 Dec 2019.

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European Journal of Sport Science, 2019
https://doi.org/10.1080/17461391.2019.1697374

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ORIGINAL ARTICLE

Workload efficiency as a new tool to describe external and internal competitive match load of a professional soccer team: A descriptive study on the relationship between pre-game training loads and relative match load

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Abstract
The current study introduces a new index for external and internal workload, "workload efficiency", and assesses in professional soccer the influence of pre-match training load on match workload efficiency. External and internal workloads were determined for 44 training sessions and 16 competitive matches using a 10-Hz global positioning system (GPS) and a 200 Hz accelerometer/heart rate monitor. Training loads were registered from day five (D-5) to day one (D-1) prior to each competitive match. Workload efficiency was calculated for each match as the ratio between overall external and internal load. A multiple stepwise regression analysis (including z-transformed variables) was used to determine training load variables that predict workload efficiency of the following matches. Training load variables of the previous days explained 26.6% of the variance in workload efficiency during the following matches. Long sprinting distance on D-3 and D-4 and total distance on D-1 positively influenced the players' workload efficiency, whereas long training durations and high training load on D-1 showed adverse effects. The present outcomes suggest that including sprint training (high sprinting distance) four and three days prior to a match, may provide a positive stimulus for the subsequent workload efficiency in matches. The negative impact of long training duration and high training load one day before the game highlights the importance of a diligent planning of the immediate competition preparation phase. This study shows that workload efficiency is a useful metric to assess match performance and that body-worn sensor technology can be useful for tailoring training loads.

KEYWORDS: Football, athlete performance, internal and external load, wearable technology, global positioning system GPS, motion tracking systems

Highlights
- Wearable sensor technology that combines motion tracking technology with heart rate monitors may provide detailed information on the external and internal workload in soccer.
- The current study introduces a new index, named workload efficiency, based on earlier concepts of internal and external workload ratios.
- The study combines internal (TRIMPMOD) and external (equivalent distance) workload parameters that specifically focus on intermittent sports.
- As a proof of concept the current study investigated whether weekly training load of a professional soccer team was related to the newly introduced index during the following matches.

Introduction

Professional soccer players are characterized by high physical fitness (Mujika, Santisteban, Impellizzeri, & Castagna, 2009). Yet physical fitness parameters established during laboratory-based tests do not necessarily determine physical performance in the match (Casajús, 2001). Defining criteria for the latter is complex as many factors, such as match importance, score line, home or away games, opponent team, recovery days, tactics and players' conscious or subconscious pacing strategies (Bradley & Noakes, 2013; Paul, Bradley, & Nassis, 2015)

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may determine match outcome. In many studies match performance was quantified by measuring absolute external loads, such as the distances covered in different speed zones or the number of high-intensity movements (Castagna, Manzi, Impellizzeri, Weston, & Barbero Alvarez, 2010; Di Salvo, Pigozzi, Gonzales-Haro, Laughlin, & de Witt, 2013; Fessi et al., 2016; Rampinini et al., 2007; Rebelo, Brito, Seabra, Oliveira, & Krustrup, 2014; Silva, Magalha, Ascensa, Seabra, & Rebelo, 2014). Yet, evaluating players' physical performance in games by solely analysing external load variables, and ignoring internal load (e.g. heart rate responses, perceived exertion) would be far too reductionist and there is a need for a combined variable that represents players overall workload during soccer games (Carling, 2013).

A possible way to deal with this issue is to express physical performance and strain as a relative variable. This relative performance variable puts the external performance or load in proportion to the respective internal load (Carling, 2013), and thus give a more comprehensive view on the players load and performance level. In this regard, Suarez-Arrones et al. (2015) proposed a basic "efficiency index" for professional soccer players in competitive games that puts the individual player's relative total distance (m/min) in proportion to the average percentage of the maximal heart rate (% HR_{max}). However, the total distance alone is a poor measure for the external load of a player, since, for example, it provides no information on accelerations or decelerations, which have substantial influence on external mechanical work (Osgnach, Poser, Bernardini, Rinaldo, & di Prampero, 2010) and match performance (Stølen, Chamari, Castagna, & Wisløff, 2005). Therefore, other external load variables (including distances in different speed zones, accelerations, decelerations and other mechanical load variables) in conjunction with the internal load parameters are needed to predict a player's current physical performance and strain (Lacome, Simpson, Broad, & Buchheit, 2018). One approach in this direction is the "equivalent distance" ED (Di Prampero, Botter, & Osgnach, 2015; Osgnach et al., 2010), which estimates from adjustable factors the distance that a player would have covered theoretically at a steady pace. Therefore, it might be valuable to establish an index based on the earlier concept of the "efficiency index" yet including several internal (e.g. HR, training impulse) and external (e.g. running distance, sprinting distance, acceleration, deceleration) training and match load parameters.

With the rise of modern tracking systems, it is nowadays very common for professional soccer clubs to monitor players' load in training sessions and games (Akenhead & Nassis, 2016). Wearable sensor technology that combines motion tracking technology with heart rate monitors provide detailed information on the external work performed by the players and the player's physiological response (i.e. internal load). Long-term seasonal monitoring enables athletes and coaching staff to better plan and to adapt the training load in order to avoid excessive fatigue and overtraining, to augment the training stimulus, to reduce injury occurrences, and to increase players physical fitness level (Akubat, 2014).

One specific challenge for athletes and coaches that can be quantitatively assessed using data from body worn-sensors is the question of how to plan training loads in the days before a competitive match. Excessive training during the week, for instance, may induce fatigue, whereas too low training intensity may induce detraining effects. Although this is a well-known concept (Clemente et al., 2019; Martín-García, Díaz, Bradley, Morera, & Casamichana, 2018; Stevens, de Ruiter, Twisk, Savelsbergh, & Beek, 2017), there are few objective criteria for optimizing training intensity in the days before a match. Consequently, professional soccer teams still adopt training strategies that predominantly evolved from practical experience.

The aims of the current study were, (a) to develop and introduce a new index, named workload efficiency, based on earlier concepts of internal and external workload ratios and (b) to analyse and describe how match workload efficiency was influenced by the training loads during the five days before the match.

Methods

Participants

Healthy male ($n = 14$; age, 22.6 ± 4.3 yrs; height, 181.5 ± 5.7 cm; weight, 75.7 ± 5.8 kg) professional soccer players of an Austrian Second League team took part in the study. Players from different field positions were investigated: central defenders ($n = 2$), wide defenders and midfielders ($n = 4$), central midfielders ($n = 4$), and forwards ($n = 4$). The tracking of training and match load is a standard procedure in the analyzed soccer club, thus this study did not cause an additional effort to the players and coaching staff and did not interfere with training and match routines in any way. The players approved the use of training and match data for the purpose of the present study by written consent. The study was carried out in conformity with the ethical standards of the declaration of Helsinki and has been approved by the Institutional Review Board of the Department of Sport Science of the University of Innsbruck.

Experimental design

Training and match load data were collected over a 13-week period during the competitive season 2017 / 2018 from March 2018 until June 2018. A total number of 44 training sessions and 16 competitive matches were tracked on both home and away grounds. This resulted in 514 individual training and 180 match observations (including data from substituted players). All training sessions were on-field training sessions where the whole team participated as well as training sessions for the non-playing squad on the day following a competitive match. Individual recovery and strength training sessions in the club's gym were not included in the data collection. All on-field training sessions consisted of a warm-up and either technical and tactical drills or on-field conditioning drills such as speed training or high-intensity interval training (HIT) drills in combination with technical and tactical training.

A typical in-season one-week micro-cycle consisted of four training days before a competitive match, followed by a regeneration training or non-starter training one day after the match and a day off two days after the match. During the season, the team regularly played two competitive matches in one week, which led to various types of micro-cycles with different numbers of training days between matches (Table I). Consequently, as a typical week was difficult to define, load analyses always included a five-day period before a competitive match – consisting of either training sessions, a competitive game or days off. For this reason, training and game loads were classified into five categories based on the number of days before game day (i.e. game day minus × day = "D−*x*"; e.g. game day minus 1 d = "D−1",).

Measurements

Data collection. The players' external and internal loads in training sessions and competitive matches were measured with the Polar Team Pro System (Polar Electro Oy, Kempele, Finland). Polar Team Pro provides portable 10 Hz global positioning (GPS) and 200 Hz movement sensors as well as heart rate sensors to track players' distance, speed, acceleration and heart rate data. Heart rate measurement was enabled by wearing specially designed underwear shirts (Polar Electro Oy, Kempele, Finland) with integrated electrodes that directly touch the skin. To avoid possible inter-unit errors (Buchheit et al., 2014), players wore the same shirt and sensors in every training and match. As reported by Giersch et al. (2018) validity and reliability of the Polar Team Pro system concerning measurement of distance and speed in linear (coefficient of variation <2.69%) and complex (changes in velocity and direction, accelerations, decelerations and stoppings) movements (coefficient of variation <1.76%) is comparable to other GPS tracking systems (Castellano, Casamichana, Calleja-González, San Román, & Ostojic, 2011; Hoppe, Baumgart, Polglaze, & Freiwald, 2018; Rampinini et al., 2015) currently used by top-level soccer clubs.

Data analysis and approach. After each training or match, GPS and heart rate data were downloaded and processed with the respective Polar Team Pro Software (Polar Electro Oy, Kempele, Finland) and spreadsheet programme (Microsoft Office Excel 2018, Microsoft Corporation, Redmond, USA). For competitive matches, the pre-game warm-up was excluded from the analyses. Incomplete or erroneous recordings were excluded from the analyses.

Variables selected for describing the external training or match load were duration (min), total distance (m), equivalent distance (m), high speed running (>14.4 km/h) distance (m), very high speed (>19.8 km/h) running distance (m), sprinting (>25.2 km/h) distance (m) as well as the number of medium (2.00–2.99 m/s²) and high (> 3.0 m/s²) accelerations and the number of medium (−2.00 to −2.99 m/s²) and high (<−3.0 m/s²) decelerations (Di Salvo et al., 2013; Fessi et al., 2016; Osgnach et al., 2010). Equivalent distance (ED) was calculated from the sessions' raw data, by applying the equations provided by Osgnach et al. (2010) and Di Prampero et al. (2015), through performing an integrated metabolic power calculation using the time course of accelerations and speed. ED is then calculated by dividing the total energy expenditure of the

Table I. Amount of team training sessions, regeneration sessions and competitive games on each loading day.

	Game day	D-1	D-2	D-3	D-4	D-5
Training sessions	0 (0%)	15 (94%)	9 (56%)	6 (38%)	10 (63%)	7 (44%)
Regeneration	0 (0%)	1 (6%)	7 (44%)	6 (38%)	1 (6%)	9 (56%)
Competitive games	16 (100)	0 (0%)	0 (0%)	4 (24%)	5 (31%)	0 (0%)

Note: Data are expressed as total number and ratio (%) of total sessions on the respective loading day.

session through the product of an adjustable terrain factor and energy cost of constant running on flat terrain. ED represents the distance that a player would have covered theoretically at a steady pace on grass using the total energy spent over the match or training (Di Prampero et al., 2015; Osgnach et al., 2010). The more energy demanding high-intensity actions such as accelerations and high-speed running players performed, the larger the ED compared to the total distance.

Variables selected for describing the internal training or match load were duration (min), Polar Training Load (PTL) (arbitrary units) and a modified training impulse (arbitrary units) specially designed for soccer players by Stagno, Thatcher, and van Someren (2007). PTL is a metrically scaled, predefined variable of the Polar Team Pro Software, which includes the session's duration and heart rate data, player's training history, a sport-specific factor as well as the player's aerobic capacity (VO_{2max} and anaerobic threshold). The modified training impulse ($TRIMP_{MOD}$) was calculated from the raw data and represents the time spent in certain heart rate zones multiplied by a weighting factor (Stagno et al., 2007).

To create the new index, named workload efficiency, which reflects the ratio of external and internal load in a match, the variable ED (Di Prampero et al., 2015; Osgnach et al., 2010) was set in relationship to the $TRIMP_{MOD}$ (Stagno et al., 2007) (i.e. ED/$TRIMP_{MOD}$). As Lacome, Simpson, and Buchheit (2018b) reported, analysing and comparing the heart rate responses in proportion to the external load of specific soccer drills represents a viable tool to describe changes in a player's workload efficiency, indicating also changes in the player's fitness level.

Statistical analyses

Statistical analysis was performed using SPSS V.24.0 (SPSS, Chicago, Illinois, USA). As external and internal load variables are highly different in magnitude, all variables were expressed in *z*-scores to make changes comparable. Z-scores of the respective variables were computed using the individual player's mean and standard deviation of every match and training day separately. A hierarchical multiple regression analysis was used to assess if and how external and internal load variables of the previous five loading days significantly predicted players' match workload efficiency. Preliminary analyses were performed to ensure there was no violation of the assumption of normality, linearity, multicollinearity and homoscedasticity. Missing values were replaced with mean values. With the whole team as sample, z-transformed workload efficiency established during the competitive matches was set as the dependent variable. All other z-transformed load variables (i.e. total distance, sprinting distance, high speed distance, very high speed distance, number of medium accelerations, number of high accelerations, number of medium decelerations, number of high deceleration, equivalent distance, PTL, $TRIMP_{MOD}$) from D-1, D-2, D-3, D-4 and D-5 were set as independent predictor variables.

Data, including internal and external load variables computed for each loading day, are presented as mean ± SD (standard deviation) and/or 95% CI (confidence interval). Statistical significance was set at $p \leq 0.05$.

Results

The newly introduced workload efficiency index during the matches was influenced by sprinting distance at D-3 ($\beta = .477$, $p < .001$), total distance at D-1 ($\beta = .380$, $p < .001$), sprinting distance at D-4 ($\beta = .312$, $p < .001$), PTL at D-1 ($\beta = -.259$, $p = .001$) and training duration at D-1 ($\beta = -.216$, $p = .007$). The five predictor variables explained 26.6% of the variance (corrected R^2 of .266, $F(5, 174) = 13.9512$, $p < .000$). Training load data show that mean internal and external load increase from D-5 over D-4 with a peak on D-3. Internal and external load are subsequently lower on D-2 and decrease to lowest scores on D-1 (Table II).

Discussion

The present study introduced the workload efficiency index and, as a proof of concept investigated whether weekly training load of a professional soccer team was related to the newly introduced index during the following matches. The study showed that match workload efficiency was predicted by training loads recorded during the days before a match, yet only to a limited extend (26.6% of the variance can be explained). The sprinting distance four and three days before a game and the total distance one day before a game seem beneficial for players' physical efficiency in the upcoming game. High training loads and long training sessions one day before a game on the other hand will negatively influence players' workload efficiency.

From a training science perspective, the training load data show that the team adopted a weekly periodization with the highest sprinting distance four and three days before a game and the lowest overall loads on D-5 and one day before game (Table II). In this context, it should be considered that the

Table II. Descriptive statistic of internal and external load variables separated by game and loading day.

	Game		D-1		D-2		D-3		D-4		D-5	
Variable	Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI
Workload efficiency (au)	47.3 (7.3)	46.3–48.4										
Duration (min)	92.2 (31.1)	87.5–96.9	81.4 (12.1)	79.5–83.2	87.2 (12.8)	54.5–89.9	91.1 (24.1)	86.6–95.5	91.6 (27.7)	88.2–95.0	80.2 (12.8)	77.5–82.9
Total distance (m)	8985.3 (2933.1)	8552.9–9415.7	4193.8 (687.6)	4089.4–4298.2	4889.0 (887.2)	4705.3–5072.8	7055.5 (2661.6)	6572.6–7538.9	5974.1 (2847.9)	5528.0–6420.2	4195.9 (702.9)	4044.3–4337.5
Equivalent distance (m)	11098.7 (3618.6)	10537.5–11601.9	5036.4 (869.4)	4904.4–5168.4	5888.9 (1164.4)	5645.0–6132.8	8707.3 (3292.4)	8109.7–9305.0	7268.3 (3554.0)	6713.1–7823.4	4984.6 (883.5)	4794.0–5175.1
High speed distance (m)	1911.3 (727.4)	1804.4–2018.3	350.9 (191.5)	321.8–380.0	543.2 (281.4)	484.3–602.2	1262.4 (795.9)	1117.9–1406.9	915.4 (853.5)	781.7–1049.1	365.5 (201.1)	322.1–408.9
Very high speed distance (m)	581.9 (299.8)	537.9–626.0	81.6 (78.3)	69.7–93.5	177.2 (146.8)	146.4–207.9	387.7 (304.6)	332.5–443.0	274.8 (299.3)	227.9–321.7	108.2 (108.8)	84.7–131.6
Sprint distance (m)	113.2 (92.3)	99.6–126.7	10.6 (21.2)	7.4–13.8	32.8 (52.1)	21.9–43.7	76.8 (85.2)	61.3–92.2	55.5 (78.8)	43.1–67.8	17.1 (37.0)	9.1–25.1
N of medium accelerations (n)	59.6 (23.6)	56.1–63.0	29.3 (9.6)	27.9–30.8	36.9 (14.4)	33.9–40.0	50.3 (18.8)	46.9–53.7	42.1 (19.1)	39.1–45.1	26.0 (9.5)	23.9–28.1
N of high accelerations (n)	11.7 (9.2)	10.4–13.1	6.2 (7.8)	5.7–6.8	7.7 (5.8)	6.4–8.9	10.2 (7.6)	8.8–11.6	9.0 (7.9)	7.8–10.2	5.3 (3.6)	4.5–6.1
N of medium decelerations (n)	59.6 (20.9)	56.5–62.7	24.7 (8.1)	23.5–25.9	30.5 (13.7)	27.7–33.3	48.2 (18.3)	44.9–51.6	38.9 (20.1)	35.8–42.0	21.6 (8.8)	29.7–23.5
N of high decelerations (n)	20.9 (10.5)	10.4–13.1	5.2 (3.6)	4.7–5.8	7.8 (6.1)	6.5–9.1	15.2 (9.9)	13.4–17.0	11.4 (9.8)	9.8–12.9	4.6 (3.6)	3.9–5.4
Polar training load (au)	262.2 (90.4)	247.9–274.5	84.3 (33.9)	79.1–89.4	101.6 (37.0)	93.8–109.3	187.6 (90.3)	171.1–203.9	157.0 (95.5)	142.1–172.0	82.4 (32.4)	75.4–89.4
$TRIMP_{MOD}$ (au)	240.2 (86.8)	227.5–253.0	54.9 (29.5)	50.5–59.4	69.4 (33.7)	62.3–76.4	159.1 (91.7)	142.5–175.8	124.5 (95.7)	109.7–139.3	54.6 (30.3)	48.1–61.1

Note: D-1, game day minus 1; D-2, game day minus 2; D-3, game day minus 3; D-4, game day minus 4; D-5, game day minus 5.

number of games on loading days D-4 and D-3 was 31% and 24%, respectively (this could partly explain the great variability in some of the external load parameters on those days). Such training load distribution and periodization strategies are in accordance with reports from the literature (Anderson et al., 2016; Stevens et al., 2017).

The finding of a negative effect of high PTL and long training duration performed one day before the match is not surprising. The high training load and training duration may induce fatigue (Thorpe et al., 2017), which negatively impacts performance (Rowell et al., 2018).

On the other hand, covering long overall running distance one day before the game, seemingly performed at a low intensity apparently has a positive effect on workload efficiency. Low intensity running in this sense could be seen as a recovery strategy (Rey, Padron-Cabo, Barcala-Furelos, Casamichana, & Romo-Perez, 2018) and thus might have a beneficial effect on the next day when the game is played. In addition to covering longer regenerative runs one day before the game data indicate that a certain weekly high-intensity load, preferentially sprint training performed 3–4 days before the game, is necessary to keep the players fit and induce workload efficiency improvements on the match day. As high-intensity bouts, like sprints, in soccer are typically characterized by relatively short recovery times (Bradley et al., 2010), it could be assumed, that the sprinting distance on those days was mainly covered in a type of sprint interval loads. In this perspective, Sloth, Sloth, Overgaard, and Dalgas (2013) reported, that there is strong evidence for an improved aerobic performance and VO_{2max} that goes in hand with an increased oxidative potential of the muscle following sprint interval training. According to Buchheit (2014), an increase in high-speed distance (above 14.4 km/h) was positively correlated to an increase in heart rate variability and therefore indicating an improvement in aerobic fitness in soccer players.

Limitations

In this study, the relative physiological load during a game was estimated with heart rate data ($TRIMP_{MOD}$) and expressed relative to an overall external performance (ED) to conclude on the individual player's workload efficiency. Interpreting the workload efficiency, it must be considered that, besides player's fitness level, heart rate could also be influenced by internal factors such as psychological stress, hydration levels and caffeine intake as well as external factors like air temperature and humidity (Carling, 2013; Paul et al., 2015). In regard to the external load measurements with GPS devices, it must be acknowledged that the validity of accelerations and distance into speed zones is acceleration and speed-dependent; that is, their validity decreases as the acceleration and speed increases (Buchheit & Simpson, 2017). These factors may have influenced ED and $TRIMP_{MOD}$ independent of the workload performed and thus the workload efficiency. Furthermore, this study provided a team average based model to assess the influence of training load variables on players' physical state. Due to non-training related factors like playing position, player's age and experience and the overall fitness level, the way and magnitude of how external load variables influence the individual's physical efficiency, can be player-specific (Lacome et al., 2018). Future studies should, therefore, aim to design individual models to provide more detailed information for the coaching staff. Lastly, it has to be stated that this study is of an observational nature and that the workload efficiency index as proposed here (i.e. for the entire team) as well as a possible player-specific one has to be validated in further studies. For now, we cannot state whether manipulating training load as suggested from this study, really will improve match performance.

Conclusion and practical applications

This study introduced a new index, named workload efficiency, based on earlier concepts yet including several internal (e.g. HR, training impulse) and external (e.g. running distance, sprinting distance, acceleration, deceleration) training and match load parameters. As a proof of concept, data show that training loads recorded during the days before a match are able to predict match workload efficiency. From a practical point of view, this study demonstrates how data from body-worn sensor technology can be evaluated to provide practical information for athletes and coaches on how to pace training load before a competitive match. Present findings suggest that for an optimal pre-game preparation, coaches should plan sprint training session during the week (3–4 days before the game) and additionally should avoid excessive training loads and long sessions one day before the game. Low intensity running the day before the game, however, can be included.

Acknowledgements

The authors thank the technical staff and the players of the professional team who participated in this study.

Disclosure statement

No potential conflict of interest was reported by the authors.

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